

Preliminary report: LLM-authored critic features for Shadow Hand control

A cautious assessment of performance, baselines, and completed controls

Executive summary

This project tests whether an LLM-authored staged program can help PPO *without entering the learned policy's action path*. The actor retains its ordinary observation and action interface; the program supplies only critic-side measurements such as stage cursor, suggested action, action/disagreement norms, and success margins. The environment reward and policy class are unchanged.

Bottom line. The current evidence establishes a narrow mechanism claim: informative authored features improve critic explained variance compared with an equally wide random-feature control. It does *not* yet establish a reliable manipulation-performance gain. Early single-seed comparisons are encouraging, whereas the strongest 12-seed control found better value fit but slightly worse one-hour graded objective and success. The scientifically defensible conclusion is that the intervention changes value prediction as intended, while the value-fit-to-policy-performance link remains unresolved.

Task and evaluation

All results below use the MJX Shadow Hand lifting a 5 cm cube with the shared PPO and evaluation stack. Graded objective is the environment's task score and success is the task success rate. Each run stores its configuration, program JSON, metric history, and final report.

Direct baseline comparison

Early direct comparison on Shadow Hand cube lift (one matched seed; 4 h/arm)

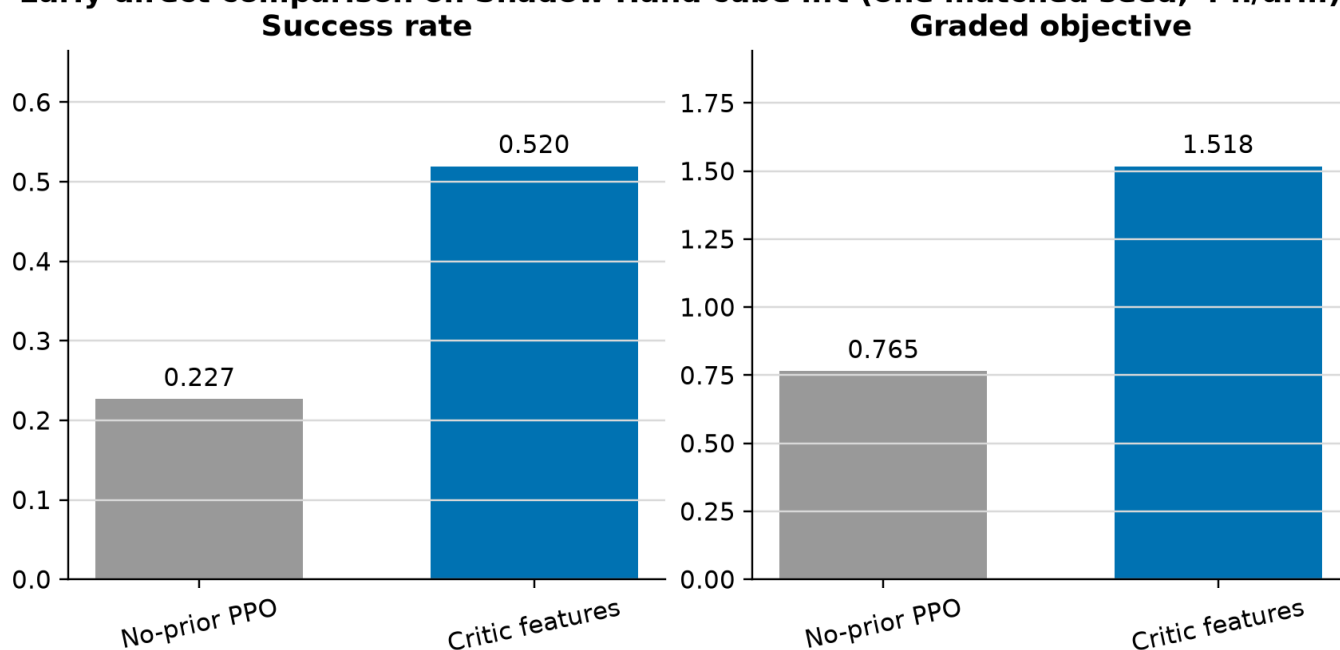


Figure 1: Early direct comparison, one matched training seed and four hours per arm. Critic features reached 0.520 success and 1.518 graded objective, versus 0.227 and 0.765 for pure PPO with no prior-derived reward or critic inputs. The setting is matched, but $n = 1$ means this is a promising lead rather than a population-level estimate.

Matched-width content control

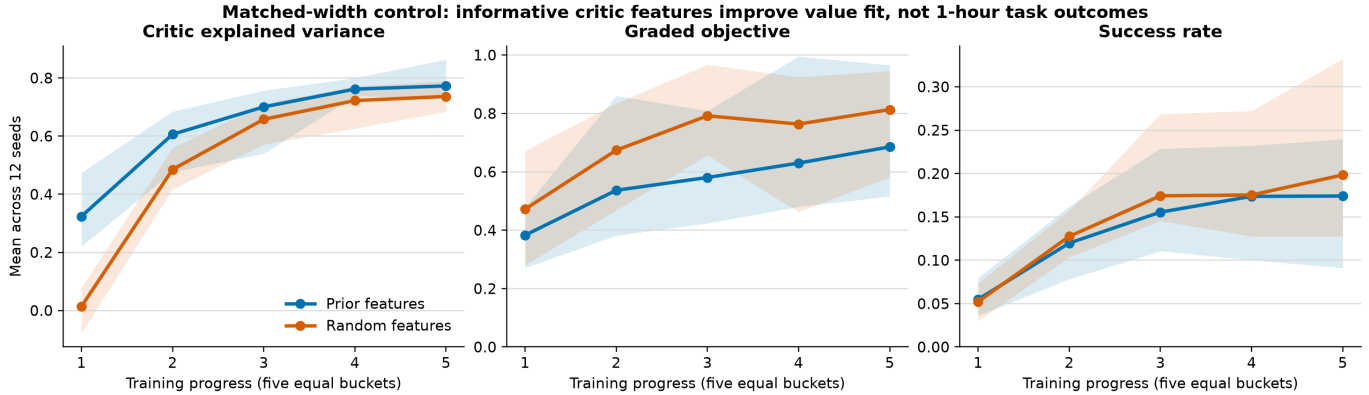


Figure 2: Rigorous content control: 12 one-hour seeds per arm. Real LLM-authored feature blocks and iid Gaussian feature blocks have identical width and PPO settings. Lines are seed means; shading spans the per-bucket seed range.

The real-feature arm finished with higher explained variance (0.773 versus 0.737), directly showing that authored content improves critic value prediction beyond an input-dimensionality effect. However, the random-feature control finished higher on graded objective (0.805 versus 0.671) and success (0.194 versus 0.174). The similar pattern in an earlier pilot means it should not be dismissed as a plotting or best-checkpoint artifact.

Completed controls and ablations

Study	Question / control	Design	Result and interpretation
Direct baseline	Critic features vs pure PPO	4 h, 1 seed	0.520 vs 0.227 success. Positive but unreplicated.
Diversity	Three-program union vs narrower representation	2 h, 1 seed; large final eval	0.458 vs 0.131 large-eval success. Supports diverse critic measurements; no training-seed replication.
Composition	Noise, simple, dual+complex	3 h, 1 seed	Noise success 0.195; real unions 0.055/0.066. More blocks were not monotonically better.
Use-information	Programs authored with vs without critic-use context	1 h, 3 seeds	Success 0.155 vs 0.126; graded 0.773 vs 0.862. Mixed result.
Matched-width content	Real union vs Gaussian features	1 h, 12 seeds	EV favors real content; one-hour behavior favors noise. Strongest causal control.

Interpretation and next experiment

The valuable result so far is a separation of mechanism from outcome. Critic-side program information is measurably predictive of return, but better in-sample value fit has not yet reliably improved short-horizon dexterous manipulation. This rules out the easy narrative that critic prediction automatically improves policy learning and points directly to the next decisive test.

Next experiment. Repeat the real-feature versus matched-noise control for four hours with at least three seeds, reporting every evaluation checkpoint rather than selected best checkpoints. A late behavioral advantage for real features would support the proposed credit-assignment pathway; continued parity or reversal would be a meaningful negative result about the limits of critic-side task structure in this PPO regime.

Run artifacts

Direct baseline: `runs/framework_cmp_4h_20260711/`
`{critic_features,no_prior_control}/`

Diversity: `runs/critic_diversity_4h_useinfo/` Composition: `runs/critic_composition_12h/`

Use-information ablation: `runs/critic_useinfo_1h_20260713/`

Matched-width control: `runs/critic_noise_vs_prior_20260721/`.